

PROJECT REPORT

1.Introduction :-

Battery Management System (BMS) is a vital electronic system for managing, controlling, and protecting rechargeable batteries. Usually, a battery pack consists of several cells interconnected together. At the same time, during charging and discharging of batteries, the condition of every cell may become different. In case such differences are not controlled, they may affect battery performance and even decrease battery life or even cause dangerous conditions.

The goal of the project is the creation of a Smart Battery Management System using ESP32 and Blynk IoT that would allow the continuous monitoring and protection of a 4-cell battery pack. As a main controller, ESP32 microcontroller is used in this project. ESP32 gets voltage readings from all four cells and also pack voltage and current information through sensor inputs.

After that, the ESP32 processes such data in order to calculate necessary battery parameters including State of Charge (SoC), cell voltage imbalance, weakest cell, strongest cell, C-rate, imbalance trend, etc. This will help in evaluating current condition of the battery.

The system also includes a fault detector module. Such faults as cell voltage out of the safe range, too big cell imbalance, pack overvoltage, too fast ADC value change, stuck sensor readings, relay mismatch are checked. In case of detecting a critical fault, the system changes operating state and turns OFF the relay in order to disconnect the battery or load.

In addition, for local battery information display, 16×2 I2C LCD display is used. There, such information as SoC, imbalance, system state, relay state, current and faults is shown. In addition, Blynk IoT platform is used for sending battery data to the cloud dashboard and, thus, remote monitoring of the system is provided.

There is also an additional feature – offline telemetry storage. In case of Internet connection interruption, the system stores telemetry data in a queue and sends the data to the Blynk server after connection recovery. In addition, a risk score, health score and maintenance recommendation are calculated in the program according to the monitored battery conditions.

As a result, this project includes many areas including embedded systems, battery monitoring, fault detection, safety control, LCD interfacing and IoT communication in one integrated system. This shows how the ESP32 microcontroller may be used for creating a smart battery management system which can be used as a prototype of modern battery-powered devices.

2.Background:-

The growing use of rechargeable batteries in electric cars, portable electronics, renewable energy storage systems, and embedded applications requires a reliable battery monitoring system and a battery protector. A battery pack typically consists of several cells that are connected together to achieve the necessary voltage and capacity. While these cells could be of the same type, their voltages, capacities, temperatures, and aging behaviour could differ while operating.

One of the issues of multi-cell battery packs is a voltage imbalance. While charging or discharging a battery, one cell could have a higher or lower voltage than others. In case such a state is persisted, the battery pack won't operate efficiently and its lifetime will be negatively impacted. That is why each individual cell should be monitored rather than the overall pack voltage.

A battery management system is used to address the above-mentioned problems by monitoring several battery parameters. It can detect abnormal voltage levels, calculate an approximate State of Charge (SoC) of the battery, measure the current and find out the difference between the highest and lowest cell voltages. Based on these values, the BMS could make a decision about normal or protected operation of the battery.

In this project, an ESP32 microcontroller is used as a central processing unit. It receives the battery measurements from the ADC inputs and evaluates them according to the programmed BMS algorithm. It detects the weakest and strongest cells, calculates cell imbalance, finds the trend of the imbalance and finds out the C-rate of the battery.

3. Learning Objectives:-

The learning objectives of this project include the following:

Understanding the functioning of Battery Management System.

Interface of sensors with ESP32.

Measuring battery cell voltages.

Calculation of SoC, C-rate and cell imbalance.

Fault detection and protection schemes.

Implementation of relay based protection system.

Use of I2C LCD display for battery information.

Connecting ESP32 with Wi-Fi and Blynk IoT.

Offline data storage and cloud telemetry.

Development of state machine based embedded system.

4. Activities and Tasks :-

There were several activities that were done while developing the Smart Battery Management System, such as design, implementation, testing and analysis. The following list describes the activities and their corresponding tasks:

1 System Planning and Design =

In order to understand what should be done and what kind of Battery Management System should be designed, it is necessary to know the requirements of such a system and the parameters which should be monitored. In general, this system was based on an ESP32 microcontroller, four cell voltage inputs, current sensor, relay, LCD display, and Blynk IoT.

2 Hardware Configuration =

In this step, the necessary GPIO and ADC inputs of ESP32 were determined and configured. Four ADC inputs were assigned for monitoring each individual battery cell. Additionally, there were other inputs and outputs for monitoring pack voltage, current sensor, relay control, and relay feedback.

3 Sensor Data Collection =

ESP32 microcontroller was programmed to collect sensor data periodically. The system collects the following values:

Voltage of all four cells

Pack voltage

Pack current

Relay feedback status

These sensor readings are pre-processed before using them for further calculations.

4 Battery Parameter Calculation =

After collecting data from sensors, it is necessary to calculate some battery parameters. The system calculates the highest and lowest cell voltage, voltage difference between them, calculates the State of Charge (SoC), calculates C-rate and detects the trend of cell imbalance.

5 Fault Detection =

There were several safety conditions which were checked in software. These are:

Cell voltage out-of-range conditions

Excessive cell imbalance

Pack overvoltage

Unexpected ADC value changes

Frozen sensor values

Relay mismatch

6 Fault State Management =

There was a state machine which was responsible for managing the current system state. The system can be in the following states: NORMAL, DEGRADED, FAILSAFE, and SHUTDOWN.

7 Protection via Relay =

The relay is used as a protection means. In normal conditions, the relay can be ON while in case of some fault conditions the relay goes OFF and switches off the battery. Feedback of relay position is also considered to ensure the relay is in the desired state.

8 Development of Display via LCD =

A 16×2 I2C LCD is added to provide display monitoring locally. Several display pages have been developed to show battery, system, relay states, current, C-rate, cell information, and fault notifications.

9 Blynk IoT Connectivity =

ESP32 has been connected to the Blynk IoT platform using Wi-Fi. Battery information has been mapped to several Blynk virtual data streams to allow the user to monitor the system remotely.

10 Offline Data Management =

Offline telemetry queue has been implemented to manage temporary loss of the Wi-Fi or Blynk connections. In case of such losses, data from the battery will be temporarily kept until connection recovery.

11 Battery Risk and Health Evaluation =

The system has been extended with a section that evaluates risk and health scores according to the battery's condition. Recommendations about battery maintenance have been also provided in order to help the user to evaluate the condition of the battery.

12 Tests and Bugs Fixing =

System logic testing included normal operation, sensors reading, fault conditions, relay control, LCD updating, and Blynk communication check. Program and system bugs were identified and fixed throughout the development process.

6. Feedback and Evidence :-

The implementation of the Smart Battery Management System offered the proof of the learned concepts of the project. Thus, the implemented code and system capabilities show how battery monitoring, battery protection, IoT communication and data analysis work in practice.

The key feedbacks and evidence received from the project include:

Battery Monitoring: The system efficiently monitors the voltage of all four cells of the battery pack, as well as the voltage and current of the entire battery pack;

Cell Analysis: The program determines the most and least powerful cells and calculates the voltage imbalance between them;

SoC Monitoring: The system estimates the SoC of the battery pack depending on the detected voltage of cells;

Fault Detection: There are different abnormal conditions, like cell imbalance, overvoltage, ADC jump, frozen sensor readings, and relay mismatch detected by the program;

Safety Protection: Relay control offers the mechanism of the battery or load disconnecting in case of any serious fault;

State Management: The NORMAL, DEGRADED, FAILSAFE, and SHUTDOWN states provide the evidence of the ability of the system to react differently in accordance with the severity of the fault;

LCD Monitoring: The 16×2 LCD displays the information about the SoC, imbalance, current, system state, relay state, and faults;

IoT Monitoring: The Blynk application enables the transmission of important battery parameters for the remote monitoring purposes;

Offline Operation: The telemetry queue is the evidence of the capability of storing the data in case of loss of network connection;

Battery Analysis: The system computes risk score, health score, and offers recommendations for further maintenance.

All in all, the feedbacks and observations during the testing's helped to check out the capabilities of the created system in terms of monitoring battery condition, identifying abnormal situations, providing the protection and sending battery information to the user.

7.Challenges & Solution :-

Challenge	Solution
Monitoring several battery cells	ADC inputs for each individual cell were created separately.
The values received from sensors may vary	Filtering was done.
Imbalance between the cells in an unsafe mode	Adaptive imbalance threshold was programmed.
Sudden failures of sensors	Jump and freeze of ADC values were detected.
Malfunction of relay	Relay's malfunctioning can be detected using its feedback.
Disconnection from Wi-Fi	Telemetry queue is used to store temporary data when connection is lost.
Restart after fault	Recovery verification takes place.
Displaying several variables on a small LCD	Several LCD pages were automatically scrolled.
Remote access	Blynk IoT platform was used by creating virtual data streams.

8. Results and Impacts :-

A successfully implemented smart battery monitoring and protection system is demonstrated here.

The major results include:

Real time monitoring of a 4 cell battery pack.

Identifying the weak and strong cells automatically.

Fault detection in batteries and sensors.

Automatic protection via relays.

Local monitoring using LCD.

Remote monitoring using Blynk.

Data storage when there is no connection to the network.

Assessment of risks and health of the battery.

Maintenance suggestions.

This system can be useful as a prototype for various applications like electric vehicles, portable battery systems, solar energy storage, and other embedded battery-powered systems.

9. Conclusion :-

Smart Battery Management System based on ESP32 and Blynk IoT proves itself as a good solution for monitoring and protection of the 4-cell battery pack. This system constantly monitors cell voltages, pack voltage, current, processes collected data and calculates such parameters as SoC, C-rate, cell imbalance, etc.

The developed fault detection system allows detecting various faults like cell imbalance, overvoltage, sensor reading abnormality, and relay mismatch. Depending on the severity of the fault the system changes its state and switches OFF the relay under severe fault conditions.

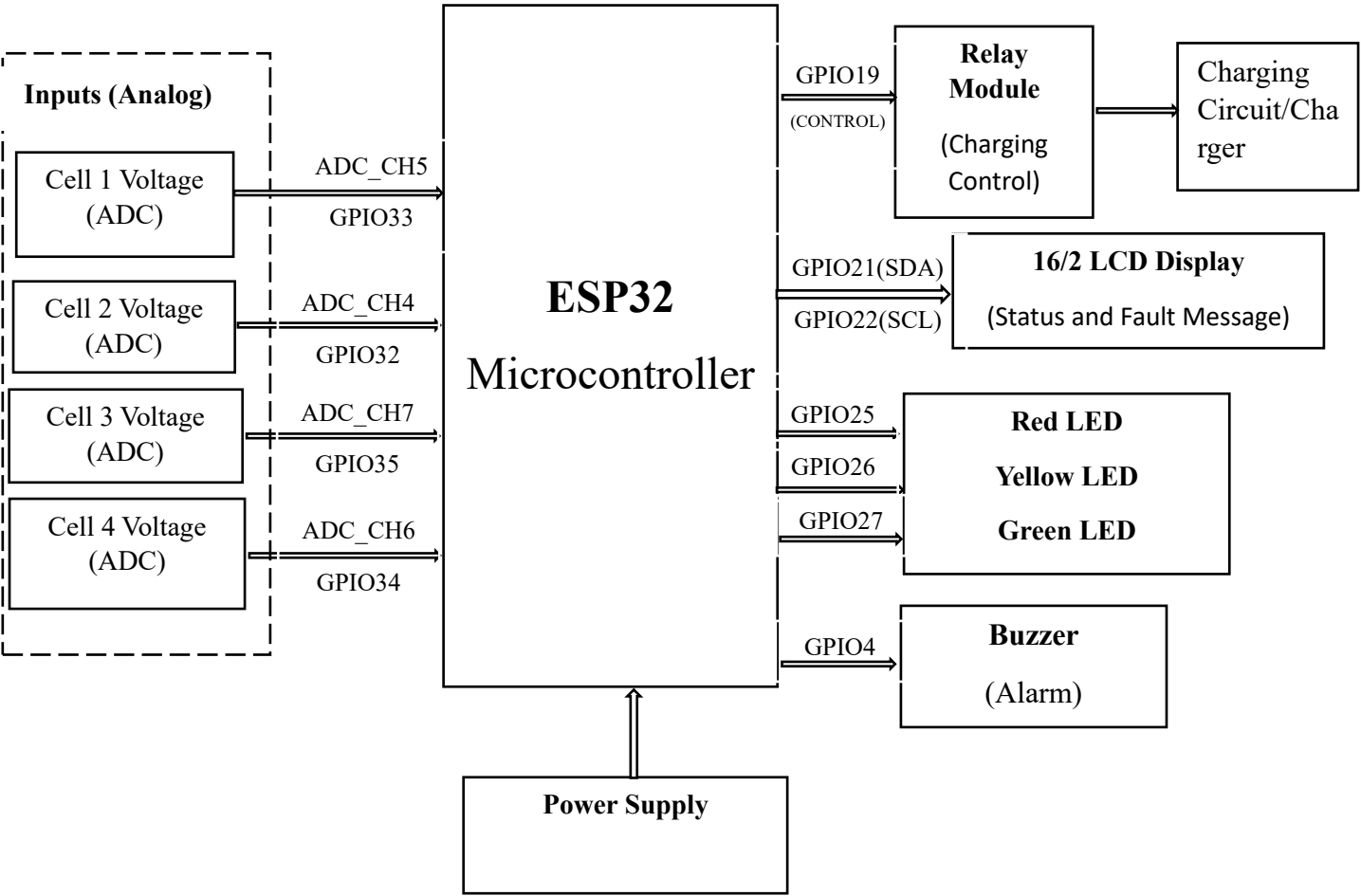
The project also implements local monitoring using 16×2 LCD and remote monitoring using Blynk IoT. Offline data queue helps to ensure telemetry in case the network connection is temporarily interrupted. Also, the risk score, health score, and maintenance recommendation features improve battery condition assessment.

All in all, the project allowed gaining useful practical experience in such fields as ESP32 programming, sensors interfacing, fault detection, relay control, LCD communication, IoT and embedded system development. This system can be improved and used for electric vehicles, energy storage system, and any other battery powered devices.

Thus, the project includes the following steps:

Monitor → Analyze → Detect Fault → Protect → Display → Communicate → Evaluate → Repeat.

System Architecture Diagram



Workflow

